

A rare PRIMER cell state in plant immunity

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Plants lack specialized and mobile immune cells. Consequently, any cell type that encounters pathogens must mount immune responses and communicate with surrounding cells for successful defence. However, the diversity, spatial organization and function of cellular immune states in pathogen-infected plants are poorly understood¹. Here we infect *Arabidopsis thaliana* leaves with bacterial pathogens that trigger or suppress immune responses and integrate time-resolved single-cell transcriptomic, epigenomic and spatial transcriptomic data to identify cell states. We describe cell-state-specific gene-regulatory logic that involves transcription factors, putative *cis*-regulatory elements and target genes associated with disease and immunity. We show that a rare cell population emerges at the nexus of immune-active hotspots, which we designate as primary immune responder (PRIMER) cells. PRIMER cells have non-canonical immune signatures, exemplified by the expression and genome accessibility of a previously uncharacterized transcription factor, GT-3A, which contributes to plant immunity against bacterial pathogens. PRIMER cells are surrounded by another cell state (bystander) that activates genes for long-distance cell-to-cell immune signalling. Together, our findings suggest that interactions between these cell states propagate immune responses across the leaf. Our molecularly defined single-cell spatiotemporal atlas provides functional and regulatory insights into immune cell states in plants.

Interactions between hosts and microorganisms are heterogeneous for multiple reasons. Multicellular host tissues are composed of diverse cell types that have distinct capacities to respond to microorganisms, and microorganisms can occupy niches heterogeneously distributed in the host. Moreover, individual interactions between cells may occur asynchronously. Thus, diverse cell states can co-exist in a tissue. Such heterogeneity can mask fundamental principles of cellular interactions when hosts and microbes are analysed at the tissue scale.

Plant–pathogen interactions have been studied to understand the molecular mechanisms that underlie host immunity and pathogen virulence². Single-cell RNA sequencing (scRNA-seq) of leaf protoplasts has revealed heterogeneous responses of plants infected by virulent bacterial and fungal pathogens^{3,4}. However, our understanding of cell-state diversity is largely limited to a specific cell type (the mesophyll³) infected by immunosuppressive virulent pathogens. Moreover, the spatiotemporal dynamics of plant immune responses are unclear owing to the low-throughput nature of transgenic reporter assays^{1,5}. Furthermore, we have a limited understanding of the gene-regulatory mechanisms that underlie cell-state diversity, such as specific binding of transcription factors (TFs) to cell-state-specific *cis*-regulatory elements (CREs). Single-nucleus assay for transposase-accessible chromatin followed by sequencing (snATAC-seq) is often used to identify potential CREs in individual cell types and states⁶, but it has not been applied to study plant immune responses. These gaps in knowledge represent substantial roadblocks to understanding how the plant immune system operates as a collective entity of cell populations with distinct functions¹.

In this study, we aimed to fill these gaps through single-nucleus multi-omic (snMultiome) and spatial transcriptomic analyses of a host plant infected by virulent or avirulent bacterial pathogens in a time-course experiment. Specifically, we use single nucleus RNA-seq (snRNA-seq), snATAC-seq and multiplexed error robust fluorescence in situ hybridization (MERFISH)⁷ and the following three bacterial pathogens: *Pseudomonas syringae* pv. *tomato* DC3000 (hereafter DC3000), DC3000 AvrRpt2 and DC3000 AvrRpm1 (hereafter AvrRpt2 and AvrRpm1, respectively). DC3000 is a virulent pathogen that can suppress plant immunity through effectors and toxins⁸, whereas AvrRpt2 and AvrRpm1 are avirulent pathogens that carry effectors indirectly recognized by plant nucleotide-binding domain and leucine-rich repeat receptors to initiate effector-triggered immunity (ETI)^{9–11}. Our single-cell multidimensional atlas reveals previously uncharacterized cell states, including a rare cell state that emerges at the centre of ETI tissue regions (designated as PRIMER cells). Another cell state surrounding PRIMER cells (designated as bystander cells) is described, and we provide insights into the functions and gene-regulatory mechanisms of these cell states.

Time-resolved snMultiome analysis

We generated a time-resolved snMultiome atlas of *A. thaliana* leaves infected by DC3000, AvrRpt2 or AvrRpm1 (Fig. 1a). Pathogens were inoculated into leaves at a low dose (optical density at 600 nm (OD₆₀₀) = 0.001), at which only a subset of plant cells are anticipated to

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